

Mohak Kapoor

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SUMMARY

Recent B.Tech graduate from New Delhi who enjoys building AI and machine learning systems end to end, from data pipelines and model development to deployment. Have built an agentic RAG assistant with a custom hybrid retrieval pipeline, and a two-stage hybrid ML system deployed live on edge hardware. Comfortable moving between classical ML, LLM-based systems, and production engineering, and always looking to learn from a strong team.

PROJECTS

CodeAtlas : Multi-Repo Agentic RAG Assistant — Python, LangChain, ChromaDB, Flask, AST, Gemini 2026

- Built an agentic RAG engine aggregating an entire GitHub portfolio by topic rather than a single repository, solving the "README domination" problem via a hybrid retrieval pipeline (dense + BM25 with cross-encoder reranking) that lifted context recall 0.48 to 0.83 over a naive baseline on a 52-case benchmark, indexed into ChromaDB using local HuggingFace embeddings for a free, private, offline-capable pipeline.
- Implemented structure-aware semantic chunking via Python AST parsing (functions and classes as discrete units) paired with Markdown header-based splitting; synced incrementally via manifest.json-tracked commit diffs to avoid full re-indexing, and served a LangChain-orchestrated Flask chat interface over Gemini with citation-backed source popups.
- Designed JudgeKit, a supplementary dual-agent LLM-as-judge eval framework (Groq/Llama 3.3 70B) scoring retrieval (Precision: 0.78, Recall: 0.83) and generation (Faithfulness: 0.88, Relevance: 0.91, Correctness: 0.82) quality, used to benchmark three retrieval architecture iterations.

Network IDS Detection Engine — Python, Autoencoder, PyTorch, scikit-learn, FastAPI, Docker Jan 2026 – May 2026

- Engineered a two-stage hybrid IDS on 2.8M rows of CICIDS2017, where unsupervised anomaly detection flags candidates for supervised classification, mirroring production platforms like Snort and Darktrace; built an end-to-end data pipeline handling ingestion, EDA, preprocessing, and temporal train/test splitting to prevent distribution leakage across unseen attack types.
- Reached 92.4% accuracy with the hybrid model across four benchmarked architectures (XGBoost F1: 0.97, PyTorch FFNN F1: 0.98, SVM F1: 0.96); Denoising Autoencoder (AUC: 0.7801) outperformed Isolation Forest on zero-day detection, and IncrementalPCA cut the feature space by 51% while retaining 98.97% variance.
- Deployed a publicly accessible real-time threat monitoring system by shipping a Dockerized FastAPI inference engine to edge hardware (Raspberry Pi 5), with a WebSocket streaming dashboard for latency metrics and Bearer token auth via Cloudflare Tunnel.

EXPERIENCE

HumanizeIQ

AI Intern

Jun 2025 – Oct 2025

Hazlet, New Jersey (Remote)

- Consolidated three manual pipeline steps into a single API call by engineering production MCP servers on Cloudflare Workers for client facing async image generation and multimodal document workflows via persistent R2 storage.
- Cut analyst turnaround from 2 hours to under 5 minutes by building LLM pipelines (GPT-4, Gemini) for automated call transcription and structured report generation, eliminating manual post-call processing.
- Supported the platform's background processing load by designing REST APIs for async job orchestration, enabling reliable long-running LLM inference within chat workflows; led daily Scrum stand-ups in an Agile environment.

ACHIEVEMENTS

- Achieved 125th rank in LeetCode Biweekly Contest 184
- Winner OSDHack'23 (JIIT)

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI, Stanford University (Nov 2024)

TECHNICAL SKILLS

Languages	Python (Advanced), TypeScript, C++
Agentic AI & LLM	LLM Integration (GPT-4, Gemini, Llama 3.3), RAG Pipelines, Agentic Systems, LangChain, Prompt Engineering
Machine Learning	Supervised Learning, SVMs, XGBoost, LightGBM, CNN, BiLSTM, Anomaly Detection, Time-Series Forecasting, Feature Engineering, Cross-Validation, Hyperparameter Tuning
Data & Retrieval	ChromaDB, Vector Databases, HuggingFace Embeddings, AST-based Chunking, Hybrid Retrieval, Pandas, NumPy, SQL
Backend & Deployment	FastAPI, Flask, REST APIs, Docker, Edge Deployment, CI/CD (GitHub Actions), MLflow
Frameworks	TensorFlow, PyTorch, scikit-learn, Hugging Face, PostgreSQL, Git

EDUCATION

Jaypee Institute of Information Technology

B.Tech in Electronics and Communication Engineering

Noida, India

2022-2026

- Relevant Coursework: Machine Learning (A+), Deep Learning (A+), Advanced Statistics (A), Queueing Theory, Computer Systems